

Blinking without eyelids: first video-documented evidence of a blink-like reflex in *Ophisops elegans* (Reptilia: Lacertidae)

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Abstract: Eyelids play a crucial role in protecting the eyes of vertebrates; however, certain reptile species, such as *Ophisops elegans*, completely lack movable eyelids. In this study, we present the first video-documented evidence of a blink-like reflex in *O. elegans*, captured through endoscopic field observations and corroborated by laboratory video analysis. The observed movement consisted of a brief convex deformation of the ocular surface, lasting approximately 0.3 s and resembling a blink despite the presence of a fixed transparent brille. This behavior was observed once in the field and twice under controlled laboratory conditions. We propose that *O. elegans* may employ subtle contractions of the ocular musculature to mimic the protective function of blinking. These findings indicate a novel compensatory mechanism in eyelidless lizards and underscore the need for further research into ocular adaptations among reptiles.

Key words: Ocular adaptation, reptile vision, eyelids, snake-eyed lizard

Blinking is a widespread and fundamental reflex observed across vertebrate species. It plays a crucial role in maintaining ocular health and function by keeping the eye hydrated and free from debris or foreign particles (Blount, 1921). In most tetrapods, blinking is accomplished through the movement of the upper and/or lower eyelids. Reptiles, in particular, exhibit a remarkable diversity in eyelid anatomy and function. While many species possess fully mobile eyelids, others employ alternative strategies for ocular maintenance. For example, some lizards, such as certain gecko species, completely lack movable eyelids and instead clean their eyes by licking them with their tongues (Masi et al., 2022).

Species of the genus *Ophisops* Ménétries, 1832, named for their snake-like immobile eyelids, are distributed from North Africa to southern India and Sri Lanka (Montgelard et al., 2020). They are also widespread across western Asia and southeastern Europe, including Türkiye (Baran et al., 2021). Members of *Ophisops* are diurnal lizards that primarily inhabit grassland habitats, with a snout-to-vent length (SVL) ranging from 35 to 75 mm (Agarwal and Ramakrishnan, 2017). They are characterized by diurnal activity, slender bodies, and a distinctive ocular anatomy in which the fused spectacle replaces movable eyelids.

Among them, *Ophisops elegans* is among the most widely distributed species, occurring throughout southeastern Europe, western Asia, and the Middle East (Tok et al., 2017; Baran et al., 2021). The species *Ophisops elegans* Ménétries, 1832, was first described from Baku, Azerbaijan (Tok et al., 2017) and has since been classified into seven subspecies: *O. e. blanfordi*, *O. e. schlueteri*, *O. e. elegans*, *O. e. ehrenbergii*, *O. e. basoglui*, *O. e. centralanatoliae*, and *O. e. macrodactylus* (Bozkurt et al., 2022). Despite its wide distribution and ecological significance, little is known about the behavioral adaptations of this species, particularly regarding mechanisms of ocular maintenance.

Within the family Lacertidae—a diverse clade comprising more than 300 species distributed across Europe, Asia, and Africa (Arnold et al., 2007)—the genus *Ophisops* represents a distinctive case. Members of this genus, including *O. elegans*, possess a fixed transparent scale covering the eye—known as a spectacle or brille—formed through the fusion of the upper and lower eyelids (Underwood, 1970; Dunayev and Orlova, 2023). Unlike most lizards, they are incapable of conventional eyelid blinking, which has led to the long-standing assumption that these species lack any blink-like behavior (Fleming, 2022; Patel et al., 2024). However, our recent observations challenge this long-held assumption.

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During a field expedition conducted in western Anatolia, Türkiye (Figure 1), we observed an unusual ocular movement in *Ophisops elegans*. Using an industrial endoscope (INSKAM G30; INSKAM, Shenzhen, China) equipped with an 8 mm lens, we recorded five individuals as they entered small rodent burrows. While most individuals exhibited no extraordinary ocular behavior, one specimen displayed a rapid and distinct deformation of the ocular surface, reminiscent of a blink. This event occurred approximately 5 s after the lizard had entered the burrow and was recorded at a rate of 30 frames per second (fps). The movement lasted approximately 0.3 s and was characterized by a brief outward bulging (convex deformation) of the brille, followed by a rapid return to its resting state (Figure 2; Supplementary video S1). To further investigate this phenomenon, eight specimens from the same locality were transferred to a controlled laboratory setting as part of another project conducted at Hacettepe University, Ankara, Türkiye. Over several days of observation, two additional instances of blink-like behavior were recorded in a single individual. These events were captured using a Samsung Galaxy S23 FE camera (Samsung Electronics, Suwon, South Korea), also at 30 fps (Figures 3 and 4; Supplementary videos S2 and S3). The observed behavior in the laboratory closely matched the blink-like motion documented in the field—characterized by a rapid, shallow deformation of the ocular surface without any associated tongue movement, head twitch, or postural adjustment.

Interestingly, this blink-like reflex was recorded in only two of the observed specimens—one in the wild and one in captivity—suggesting that it may represent a rare behavior or a highly context-dependent response. Notably, all instances of the movement occurred shortly after the animal entered a confined or low-light space, such as a

tunnel or the corner of an enclosed tank. This correlation raises the possibility that the blink-like response is triggered by environmental cues such as spatial constriction, reduced illumination, or altered sensory input.

The diversity of eyelid configurations within *Ophisops* suggests that this genus may serve as an important model for understanding the evolutionary transition from mobile eyelids to fixed spectacles (Patel and Vyas, 2020; Patel et al., 2024). In this context, the blink-like reflex of *O. elegans* may represent either a retained ancestral trait or a compensatory adaptation associated with the evolution of the spectacle (Underwood, 1970; Schwenk, 2000). Comparative studies across lacertid lizards and snakes may therefore yield valuable insights into the evolutionary drivers and functional consequences of eyelid reduction and spectacle development.

Ophisops elegans possesses transparent eyelids fused into a spectacle, resembling those of snakes. Dunayev and Orlova (2023) noted that the spectacle of *O. elegans* may retain a degree of residual mobility, suggesting the possibility of blink-like movements. However, no direct evidence of such behavior had been reported prior to the present study. Our observations therefore constitute the first video-documented evidence of a blink-like reflex in this species. These observations demonstrate that *O. elegans* possesses a blink-like reflex and open new avenues for investigating the ocular biology of reptiles with fixed spectacles. We hypothesize that *O. elegans* may possess superficial extraocular muscles capable of producing subtle contractions in the tissues surrounding or underlying the brille. Such contractions could momentarily distort the spectacle surface, producing a blink-like effect that may clean the eye or provide brief protection against mechanical stress or particulate intrusion—functions typically performed by conventional eyelid blinking in other species (Stevens, 2013).

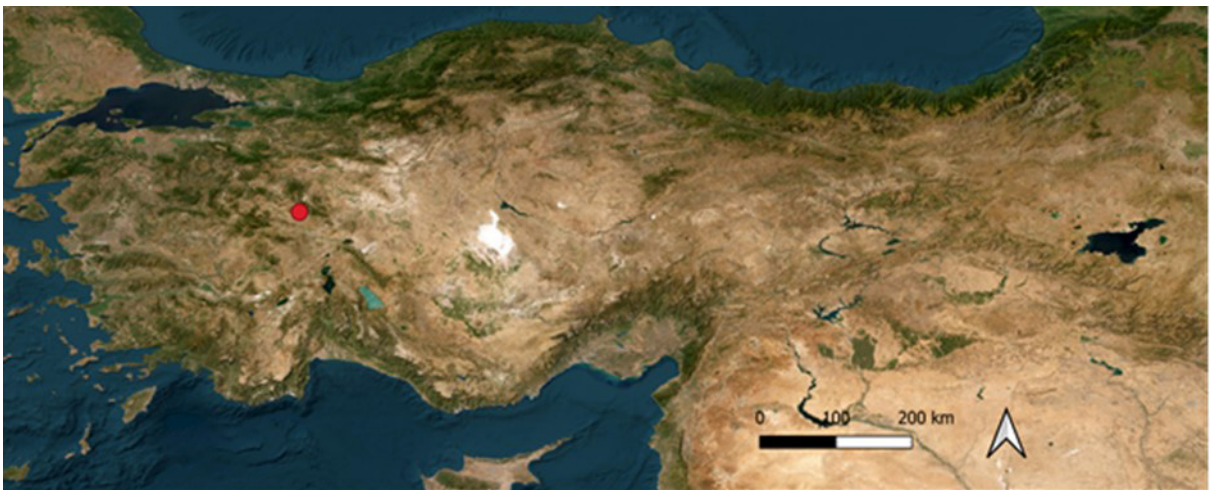


Figure 1. Field location of the *Ophisops elegans macrodactylus* observation in western Anatolia, Türkiye.



Figure 2. Sequential images captured from a video recorded on 24 June 2025 at 17:24, illustrating the blink-like behavior in *Ophisops elegans macrodactylus* (see Supplementary video S1): (A) before the blink-like movement; (B) and (C) during the blink-like movement; (D) after the blink-like movement.

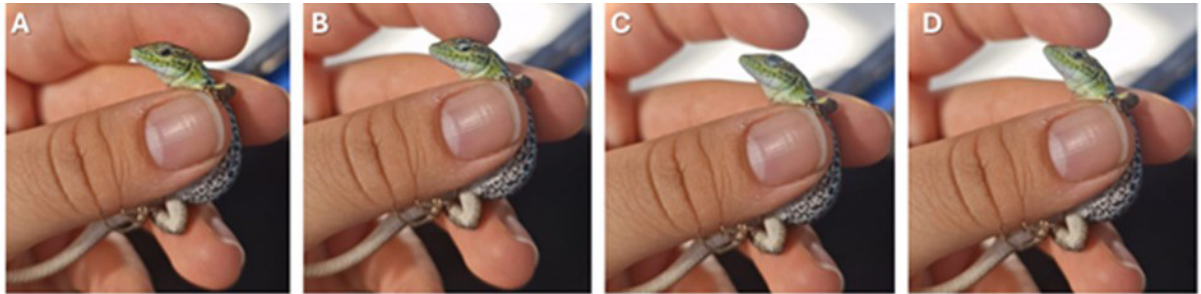


Figure 3. Sequential frames captured from a video recorded on 3 July 2025 at 16:00, showing the blink-like movement observed in *Ophisops elegans macrodactylus* (see Supplementary video S2): (A) before the blink-like movement; (B) and (C) during the blink-like movement; (D) after the blink-like movement.



Figure 4. Sequential frames captured from a video recorded on 3 July 2025 at 15:26, showing the blink-like movement observed in *Ophisops elegans macrodactylus* (see Supplementary video S3): (A) before the blink-like movement; (B) during the blink-like movement; (C) after the blink-like movement.

The presence of this reflex may indicate an adaptive or compensatory mechanism associated with the evolutionary loss of eyelid mobility. Comparative data from congeners reveal striking variation: species such as *O. leschenaultii* and *O. nictans* retain distinct eyelids, whereas *O. beddomei* and *O. venustus* exhibit partial eyelid functionality and a blink-like reflex (Patel and Vyas, 2020; Patel et al., 2024). This diversity suggests an evolutionary continuum within the genus, with the blink-like behavior of *O. elegans* potentially representing a retained or modified trait that confers selective advantages under specific ecological conditions.

Further research—including histological examinations of the ocular musculature and biomechanical analyses of spectacle deformation—is required to confirm the mechanism underlying this behavior. Additionally, a broader comparative survey across related species could help determine whether this reflex is unique to *O. elegans*

or more widespread among spectacle-bearing lizards. In conclusion, we present the first video-documented evidence of a blink-like reflex in *Ophisops elegans*, a species previously believed to be incapable of such ocular movement owing to its fused eyelids. Our findings, based on both field and laboratory recordings, suggest that this behavior involves a rapid deformation of the brille, potentially driven by underlying muscular activity. Future work, including controlled experiments on environmental triggers (e.g., reduced illumination and spatial constraints) and histological analyses of spectacle-associated musculature, will be essential to elucidate the proximate causes and functional significance of this reflex.

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Ethical approval

Fieldwork was conducted under permit no. E-21264211-288.04-13202316, issued by the Republic of Türkiye Ministry of Agriculture and Forestry, General Directorate of Nature Conservation and National Parks (Ankara, Türkiye), and permit no. 52338575-15, granted by the Animal Research Ethics Committee of Hacettepe University, Ankara, Türkiye.

Conflict of interest

The authors declare that they have no conflict of interest.

Author contributions

TMÇ, YÇ, TK, ACK, MK, and MKŞ conceived the study (TMÇ, YÇ, TK, ACK, and MK contributed equally; MKŞ led). TMÇ and MKŞ curated the data (MKŞ led). All authors contributed equally to the investigation. TMÇ

and MKŞ jointly developed the methodology. ACK and MK performed validation (equal). TMÇ, YÇ, TK, and MK prepared the visualizations (equal). MKŞ provided resources. The original draft was prepared by MKŞ with equal contributions from TMÇ, ACK, and MK. All authors contributed to the review and editing, led by MKŞ. All authors read and approved the final manuscript.

Declaration of generative AI and AI-assisted technologies

During the preparation of this work, the authors used the Open AI GPT-4o model (OpenAI, San Francisco, CA, USA) for language proofreading. After using this tool, the authors reviewed and edited the content as necessary and take full responsibility for the final version of the manuscript.

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Supplementary videos

Supplementary videos S1–S3 (high-frame-rate recordings of blink-like events in *O. elegans*) are available at the following link: <https://aperta.ulakbim.gov.tr/records/286712>